

$$2G + 2Gu = 3k$$

$$2G(1+u) = 3k - 6ku$$

$$2G(1+u) = 3k(1-2u)$$

$$\frac{2G}{3k} = \frac{(1-2u)}{(1+u)}$$



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN
CIVIL ENGINEERING

ECV 203: STRENGTH OF MATERIALS I

2ND SEMESTER 2020/2021

TIME: 2 HOURS

INSTRUCTIONS:

Attempt Question ONE and any TWO questions

QUESTION ONE (30 MARKS)

- Define principal planes (2 marks)
- Given $\sigma_1 = 90 \text{ N/mm}^2$; $\sigma_2 = -60 \text{ N/mm}^2$; $\tau_{xy} = 20 \text{ N/mm}^2$. Find the principal stresses and maximum shear stresses, and their orientation. (10 Marks)
- Calculate the modulus of rigidity and bulk modulus of a cylindrical bar of diameter 30 mm and of length 1.5 m if the longitudinal strain in a bar during tensile stress is four times the lateral strain. Find the change in volume, when the bar is subjected to a hydrostatic pressure of 100 N/mm^2 . Take $E = 1 \times 10^5 \text{ N/mm}^2$ (8 Marks)
- For the beam shown in figure 1, calculate the
 - maximum bending stress
 - shear stress at a distance of 50 mm above the neutral axis. (10 marks)

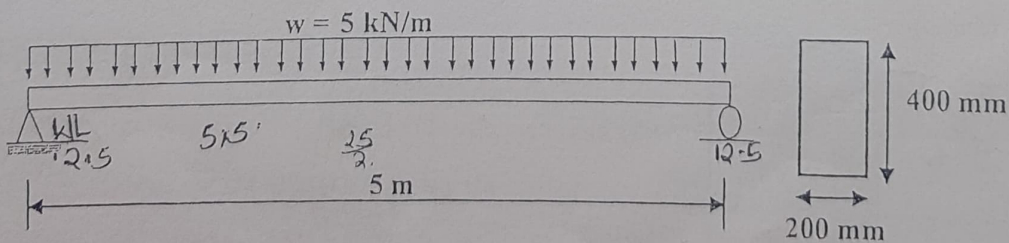


Figure 1

QUESTION TWO (20 MARKS)

- a) Evaluate the principal stresses and principal planes for the state of stress shown in Figure 2

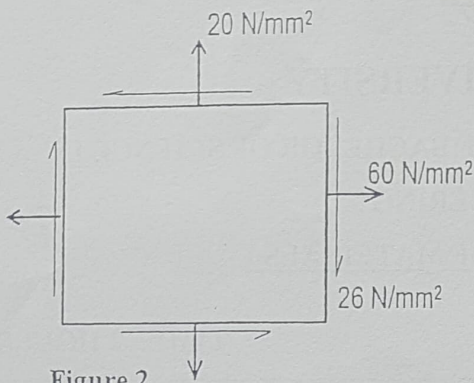


Figure 2

- b) The state of stress at a point in a loaded solid is prescribed on two faces of an element whose shape is a triangular prism as shown in figure 3. Evaluate the principal stresses and principal planes.

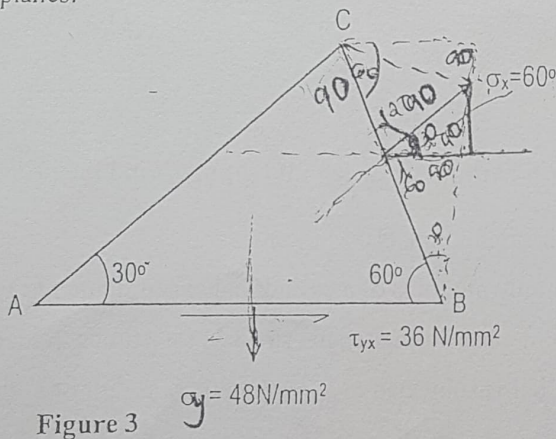


Figure 3

QUESTION THREE (20 MARKS)

- a) The I-beam shown in figure 4 is subjected to shear force $V = 5 \text{ kN}$. The flange and the web both have thickness of 20 mm, the height of the beam is 150 mm and the width is 100 mm. The beam has a moment of inertia of $I = 5.2 \times 10^4 \text{ N/mm}^3$. Determine the shear stress at a-a' and b-b'. (5 Marks)

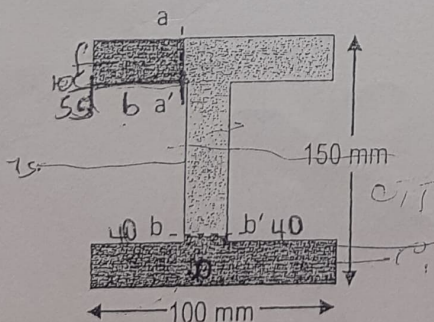


Figure 4

$$\frac{P}{k \cdot t}$$

$$\frac{5000 \text{ N} \times 40 \times 20 \times 65}{5.2 \times 10^4 \times 40} =$$

- b) Determine the centroid of the section shown in figure 5 (5 Marks)

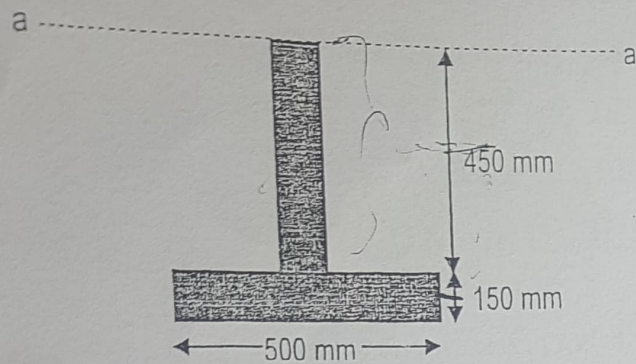


Figure 5

- c) The steel flat plate AB of 1 cm thickness shown in figure 6 tapers uniformly from 10 cm to 5 cm width in a length of 40 cm. From first principles, determine the elongation of the plate if an axial tensile force of 5000 kg acts on it. Take $E = 2.0 \times 10^6 \text{ Kg/cm}^2$. (10 Marks)

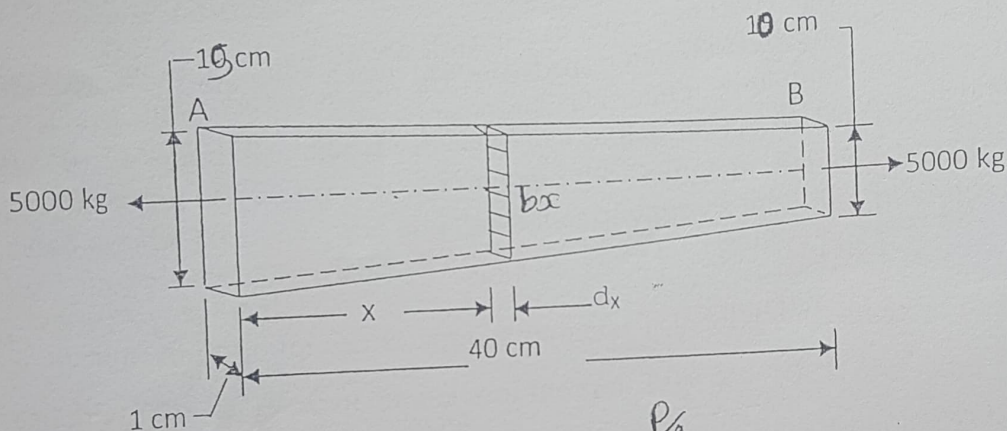


Figure 6

$$\frac{P/A}{L} = \epsilon \quad \frac{P/A}{L} \cdot L \cdot dx = \epsilon \cdot dx$$

* QUESTION FOUR (20 MARKS)

A beam of rectangular section 100 mm wide and 300 mm deep is subjected to a bending moment of 70 kNm and shear force of 100 kN. Find the principal stresses at 0, 50, 100 and 150 mm above the neutral axis. (20 Marks)

* QUESTION FIVE (20 MARKS)

Discuss the 5 failure theories for ductile and brittle materials. (20 Marks)



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN CIVIL ENGINEERING

ECV 203: STRENGTH OF MATERIALS

DATE: Monday, 3rd February 2020

TIME: 8.00a.m-10.00a.m

INSTRUCTIONS:

Attempt Question ONE and any other TWO Questions

QUESTION ONE (30 MARKS)

- a) State clearly the Hooke's law. (2 marks)
- b) A bar of 30 mm diameter is subjected to a pull of 60 kN. The measured extension on gauge length of 200 mm is 0.09 mm and the change in diameter is 0.0039 mm. calculate the Poisson's ratio and the values of young's modulus, modulus of rigidity and bulk modulus. (13 Marks)

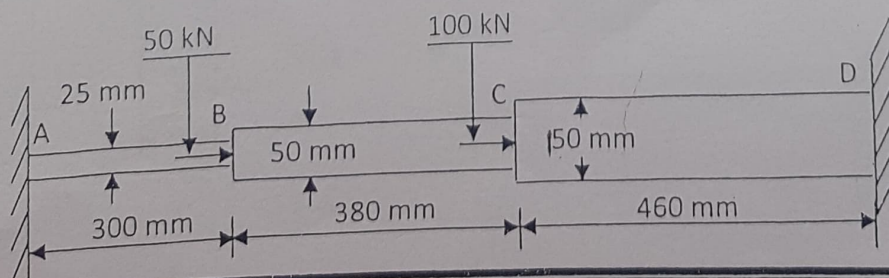
$E = 2G(1 + \mu)$
 $E = 3K(1 - 2\mu)$
A circular aluminium rod 10 mm in diameter is loaded with an axial force of 2 kN. Calculate the decrease in diameter of the rod. Take $E = 70 \text{ GN/m}^2$ and poisson's ratio, $\mu = 0.33$ (6 Marks) $1.2 \times 10^{-3} \text{ mm}$.

c) Given $\sigma_x = 3 \text{ MPa}$; $\sigma_y = 5 \text{ MPa}$; $\tau_{xy} = 3 \text{ MPa}$. Find the principal stresses and their orientation. (5 Marks)

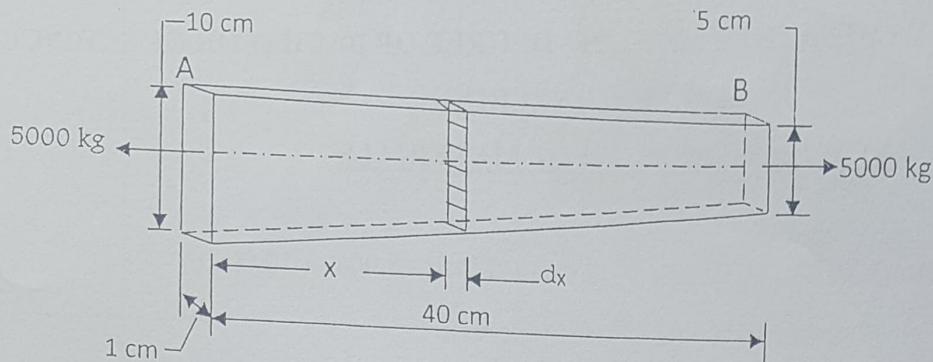
d) A 100 X 150 mm wooden cantilever beam is 2m long. It is loaded at its tip with a 4 kN load. Find the maximum bending stress in the beam. (4 Marks)

QUESTION TWO (20 MARKS)

- a) A circular steel bar ABCD rigidly fixed at A and D is subjected to axial load of 50 kN and 100 kN at B and C as shown below. Find the loads shared by each part of the bar and the displacement of the points B and C. take E for steel as 205 kN/mm^2 . (10 Marks)



- b) A steel flat plate AB of 1 cm thickness tapers uniformly from 10 cm to 5 cm width in a length of 40 cm. From first principles, determine the elongation of the plate if an axial tensile force of 5000 kg acts on it. Take $E = 2.0 \times 10^6 \text{ Kg/cm}^2$. (10 Marks)



QUESTION THREE (20 MARKS)

A 150X250 mm rectangular beam section is subjected to a bending moment of 100 kNm and shear force of 120 kN. Find the principal stresses at 0, 40, 80 and 120mm above the neutral axis. (20 Marks)

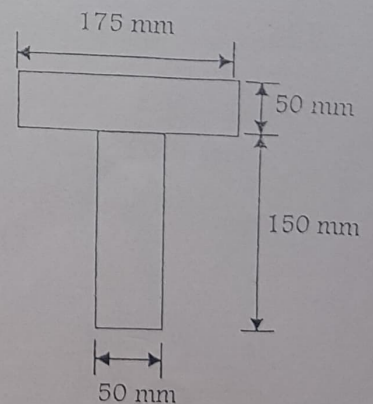
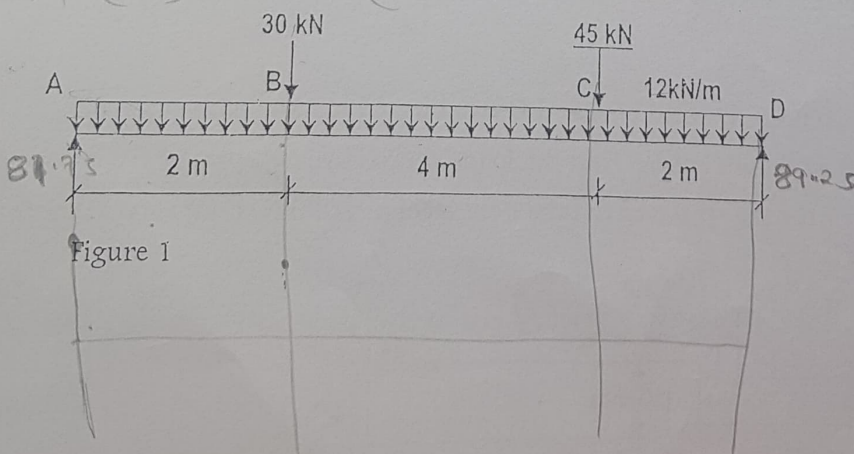
QUESTION FOUR (20 MARKS)

- a) A beam has a hollow circular cross section 40 mm outer diameter and 30 mm inner diameter. It is made from metal with a modulus of elasticity of 205 GPa. The maximum tensile stress in the beam must not exceed 350 MPa. Calculate the following

- the maximum allowable bending moment
- the radius of curvature (4 Marks)

- b) Determine the maximum bending stress in beam AB as shown in the figure and draw the stress distribution over the cross section at this location (16 Marks)

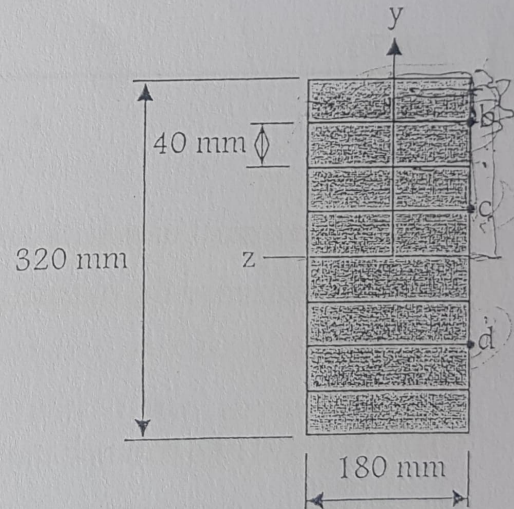
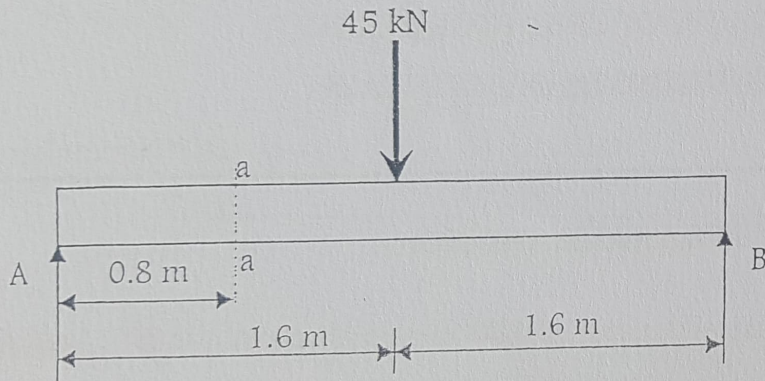
$\sum 30 + 45 = R_A + R_D$
 $\sum M_A = (30 \times 2) + (45 \times 6) - 8 R_D = 0 \Rightarrow (12 \times 8 \times 4) = 0$



QUESTION FIVE (20 MARKS)

A 3.2 m long simply supported laminated wood beam consists of eight 40 mm by 180 mm planks glued together to form a section 180 mm wide by 320 mm deep, as shown. The beam carries a 45 kN concentrated load at mid-span. At section a-a located 0.8 m from support A, determine

- the average shear stress in the glue joints at b, c, and d
- the maximum average shear stress in the cross section.





ENG. 018090
CLASS OF 2017

KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(CIVIL ENGINEERING)

ECV 203: STRENGTH OF MATERIALS 1

DATE: Monday, 18th March 2019

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

Attempt Question ONE and any TWO Questions

QUESTION ONE (30 MARKS)

- a) State clearly the Hooke's law. (2 marks)
- b) Find the moment of inertia about X-X axis and Y-Y axis of the angle section in figure 1 below. (18 Marks)

force required to deform a spring linearly is directly proportional to the distance.
 $F = k \cdot x$
force spring constant

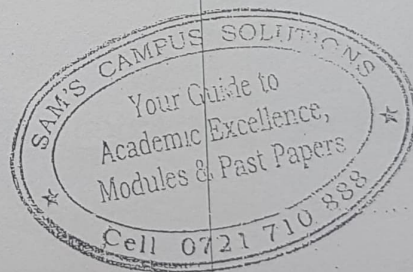
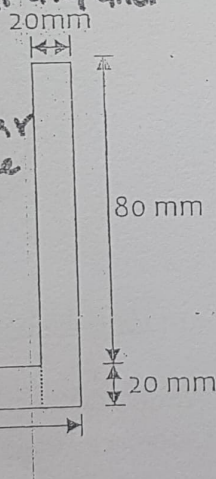
$$I_{xx} = I_g + ah^2 \rightarrow h = 5$$

Moment = area $\times$ perpendicular distance
 $M = A \times Y$

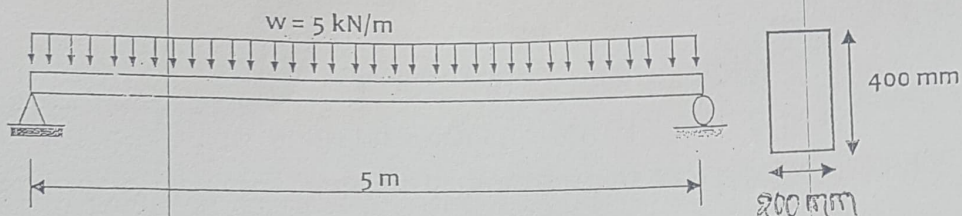
Moment of inertia = moment $\times$ perpendicular distance
 $I_{xy} = A \cdot Y \cdot x$
 $= AY^2$

$$I_{yy} = A \cdot x^2 + x$$
$$= Ax^2$$

Figure 1



- c) Calculate the maximum bending stress for the beam shown in figure 2. (5 marks)



- d) For the given state of stress shown in Figure 3, determine the normal and shearing stresses exerted on the oblique face of the shaded triangular element shown. Use a method of analysis based on the equilibrium of the element. Clearly show the diagrammatic representation for the stresses, areas and forces. (5 Marks)

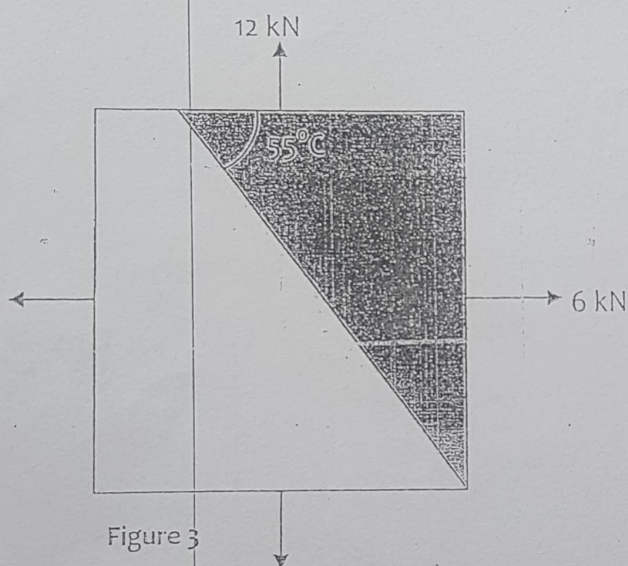
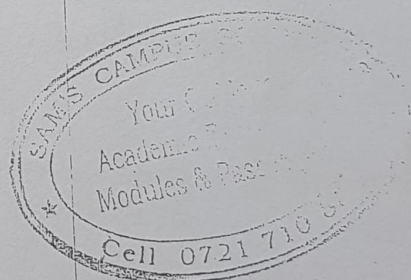


Figure 3



QUESTION TWO (20 MARKS)

- a) A mild steel rod of 20 mm diameter and 300 mm long is enclosed centrally inside a hollow copper tube of external diameter 30 mm and internal diameter 25 mm as shown in Figure 4. The ends of the rod and tube are brazed together, and the composite bar is subjected to an axial pull of 40 kN. E for steel and copper is 200 GN/m^2 and 100 GN/m^2 respectively.
- Find the stresses developed in the rod and the tube. (8 Marks)
 - Find the extension of the rod. (2 Marks)

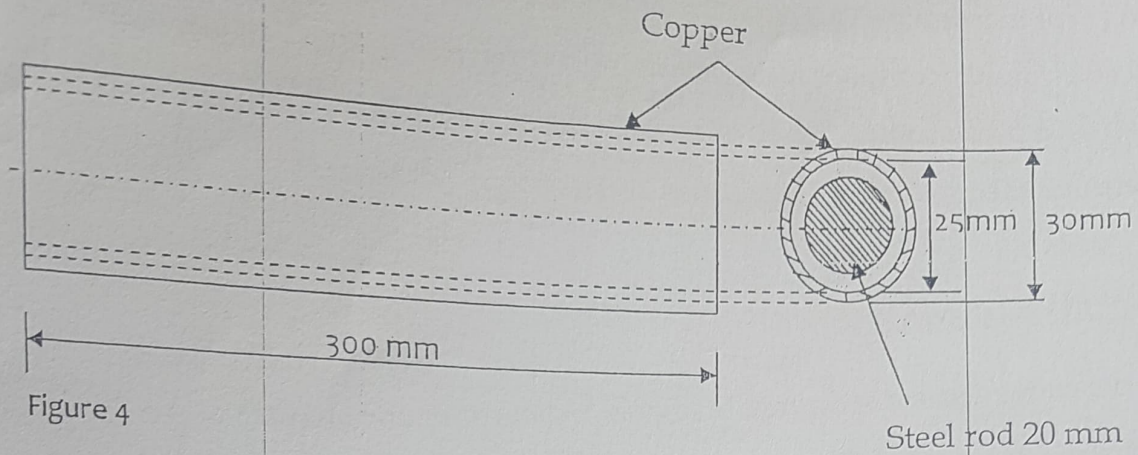


Figure 4

- b) A steel flat plate AB of 1 cm thickness tapers uniformly from 10 cm to 5 cm width in a length of 40 cm (Figure 5). From first principles, determine the elongation of the plate if an axial tensile force of 5000 kg acts on it. Take $E = 2.0 \times 10^6 \text{ Kg/cm}^2$. (10 Marks)

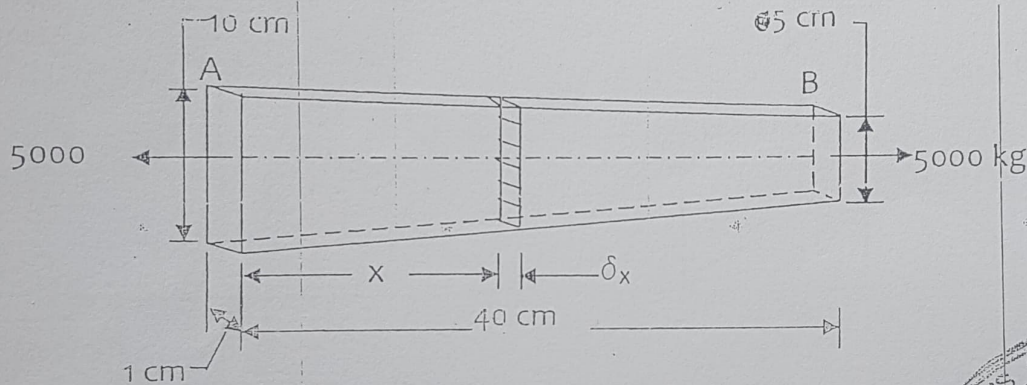


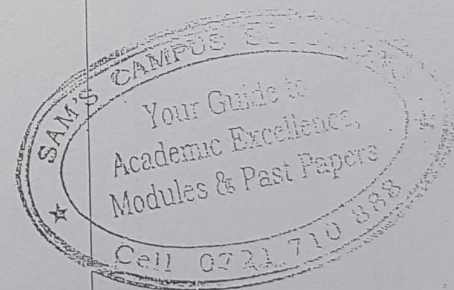
Figure 5

QUESTION THREE (20 MARKS)

A beam of rectangular section 100 mm wide and 240 mm deep is subjected to a bending moment of 76.8 kNm and shear force of 96 kN. Find the principal stresses at 0, 40, 80 and 120 mm above the neutral axis. (20 Marks)

QUESTION FOUR (20 MARKS)

- a) A bar of 30 mm diameter is subjected to a pull of 60 kN. The measured extension on gauge length of 200 mm is 0.09 mm and the change in diameter is 0.0039 mm. calculate the Poisson's ratio and the values of young's modulus, modulus of rigidity and bulk modulus. (10 Marks)



- b) At a point in a strained material the principal stresses are 150 N/mm^2 (tension) and 80 N/mm^2 (compression). Determine normal stress, shear stress, resultant stress on a plane inclined at 50° to the axis of major principal stress. Also determine the maximum shear stress at that point. (10 Marks)

QUESTION FIVE (20 MARKS)

- a) For the state of stress indicated in Figure 6, determine the stresses at $\theta = 45^\circ$ (5

Marks)

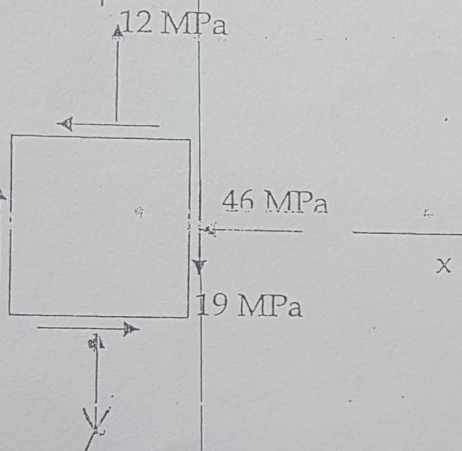
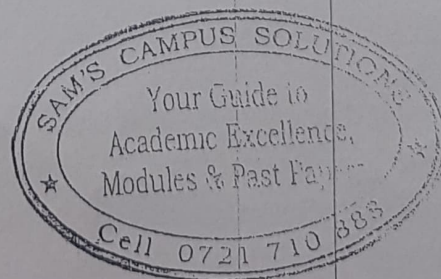


Figure 6

- b) Given that $\sigma_x = 84 \text{ MPa}$, $\sigma_y = -30 \text{ MPa}$ and $\tau_{xy} = -32 \text{ MPa}$ determine the principal stresses and maximum shear stress and their directions. (15 Marks)





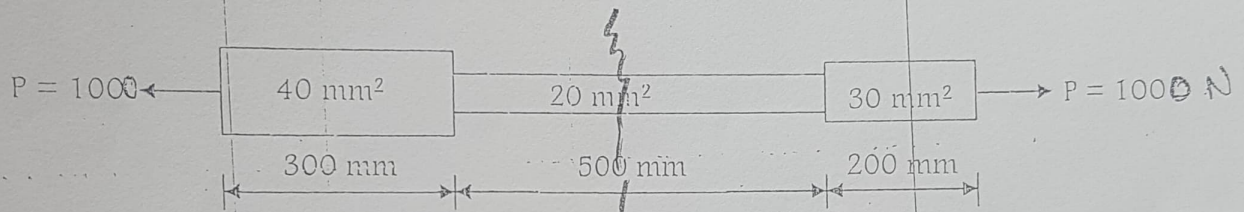
KENYATTA UNIVERSITY
DEPARTMENT OF CIVIL ENGINEERING
ECV 203: STRENGTH OF MATERIALS I

CAT I

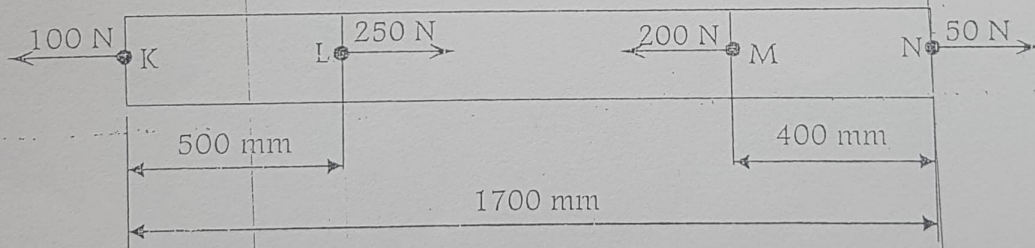
TIME: 2 HOURS

Attempt all the questions

- 1) A composite load is 1000 mm long, its two ends are 40 mm² and in 30 mm² area and length are 300 mm and 200 mm respectively. The middle portion of the load is 20 mm² in area and 500 mm long. If the rod is subjected to an axial tensile load of 1000 N, find its total elongation. ($E = 200 \text{ GPa}$)



- 2) The figure below shows a steel rod of 25 mm² cross sectional area. It is loaded at four points, K, L, M and N. Find its total elongation. ($E = 200 \text{ GPa}$)



- 3) A circular bar 40 mm diameter carries an axial tensile load of 105 kN. Find the value of shear stress on the planes on which the normal stress has a value of 50 MN/m² tensile.





KENYATTA UNIVERSITY
DEPARTMENT OF CIVIL ENGINEERING
ECV 203: STRENGTH OF MATERIALS I

C.A.T I

26th November, 2019

Attempt all the questions

- a) A hollow steel tube with an inside diameter of 100 mm must carry a tensile load of 400 kN. Determine the outside diameter of the tube if the stress is limited to 120 MN/m (5 Marks)
- b) At a point in a strained material the principal stresses are 150 N/mm² (tension) and 80 N/mm² (compression). Determine normal stress, shear stress, resultant stress on a plane inclined at 50 degrees to the axis of major principal stress. Also determine the maximum shear stress at that point. (10 Marks)

$$\sigma = 120$$



C.A.T.I

2020/2022

Time: 30 Minutes

Attempt all the question

A composite bar made up of aluminium bar and steel bar is firmly held between two unyielding supports as shown below. An axial load of 20 tonnes is applied at B.

- Find the stresses in each material.
- If a force applied at the ends of the composite bar causes a contraction of 0.02085 cm, compute stresses in each material.

Take E for aluminium and steel as $0.7 \times 10^6 \text{ kg/cm}^2$ and $2.1 \times 10^6 \text{ kg/cm}^2$ respectively.

